DLP Assignment 01 — Report

Object Detection · Semantic Segmentation · Recurrent Networks

Field	Your entry
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Course / section	DLP – 7B
Submission date	8/12/2026
Hardware used	Colab T4
Total GPU time	3 hours
Random seed	42

0. Environment and reproducibility

```
!pip -q install torch torchvision ultralytics segmentation-models-pytorch torchmetrics pycocotools oper
!python env_check.py

Python  : 3.13.16  (Linux-6.6.159+-x86_64-with-glibc2.39)
[  OK   ] torch                          2.11.0+cu130
[  OK   ] torchvision                    0.26.0+cu130
Creating new Ultralytics Settings v0.0.8 file ✅
View Ultralytics Settings with 'yolo settings' or at '/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see
[  OK   ] ultralytics                    8.4.203
[  OK   ] segmentation_models_pytorch    0.6.0
[  OK   ] torchmetrics                   1.9.0
[  OK   ] pycocotools                    ?
[  OK   ] cv2                            4.14.0
[  OK   ] matplotlib                     3.10.0
[  OK   ] sklearn                        1.6.1
[  OK   ] pandas                         2.2.3
[  OK   ] tqdm                           4.67.3

CUDA available : True
GPU            : Tesla T4
GPU memory     : 14.6 GiB

Environment looks good. Start with setup/download_datasets.py
```

1. Question 1 — Object detection: R-CNN family vs YOLO [35 marks]

1.1 The detection split

```
[15]
✓ 1m
!python setup/voc_to_yolo.py

train images : 2000
val images   : 500
objects per class (whole set):
   person     3655
   car        948
   dog        421
   cat        266
   bicycle    284

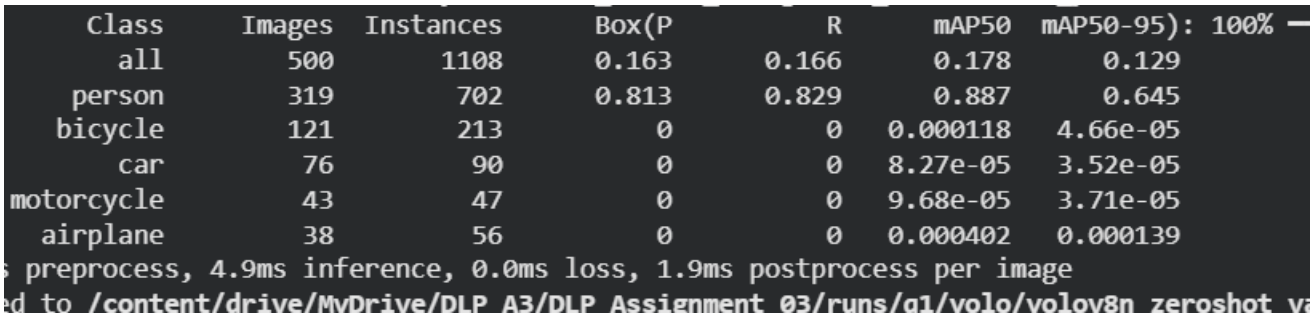
data.yaml written to /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/data/voc_yolo/data.yaml
Copy the per-class counts into your report - the class imbalance here explains
a lot about the per-class AP numbers you are about to see.
```

Class	Percentage
Person	65.6%
Car	17.0%
Dog	7.5%
Bicycle	5.1%
Cat	4.8%

Yolo Label format: <class_id> <x_center> <y_center> <width> <height>

Class imbalance: Long tail class imbalance: person class has 3655 instances whereas minority classes such as cat and bicycles accounts for a small percentage. During training, loss gradients will be dominated by person and car leading both detectors to achieve higher detection confidence and recall on person while struggling with underrepresented classes. This will result in lower per call average precision for minority categories.

1.2 Zero-shot COCO baseline



Class	Images	Instances	Box(P	R	mAP50	mAP50-95): 100%
all	500	1108	0.163	0.166	0.178	0.129
person	319	702	0.813	0.829	0.887	0.645
bicycle	121	213	0	0	0.000118	4.66e-05
car	76	90	0	0	8.27e-05	3.52e-05
motorcycle	43	47	0	0	9.68e-05	3.71e-05
airplane	38	56	0	0	0.000402	0.000139

preprocess, 4.9ms inference, 0.0ms loss, 1.9ms postprocess per image
d to /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q1/yolo/yolov8n_zeroshot_v

Baseline metric:

- mAP@0.5 : 0.178
- mAP@0.5:0.95 : 0.129
- Inference speed: 4.9 ms/image

Why mAP at 0.5 is far above zero (17.8%):

Yolov8n checkpoint was pre-trained on COCO, which already contains person class at index 0. Because person represents 65.6% of all objects in our dataset, the model achieves 88.7% mAP 0.5 on person right out of the box, pulling overall average score above zero.

Why it is not a clean measure of zero-shot ability:

- **Class index mismatch:** COCO default indices for 0 to 4 are person, bicycle, car, motorcycle, airplane. However, our 5-class target layout is person, car, dog, cat, bicycle. Without fine tuning the head, index 2(dog) was incorrectly evaluated against coco's car, index 3 against motorcycle, and index 4 against airplane. This label mis alignment caused near zero Aps for 4 out of the 5 classes.
- **Not true Zero-Shot:** Network was explicitly trained on these exact oject classes during COCO pretraining. Therefore this experiment measures off-the-shelf pre-trained feature transfer not true zero-shot generalization to unseen concepts.

1.3 Fine-tuned YOLOv8n

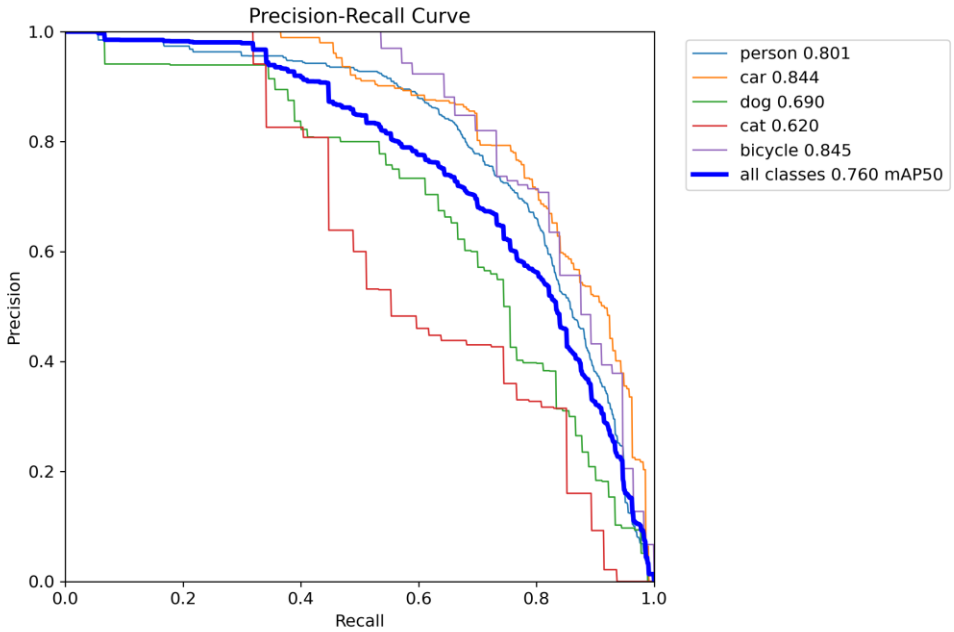
Overall performance metrics:

Metric	Value
m AP 0.5	0.754
m AP 0.5:0.95	0.4975
Precision	0.704
Recall	0.682
Inference latency	4.7 ms/image
Throughput	212.8 FPS

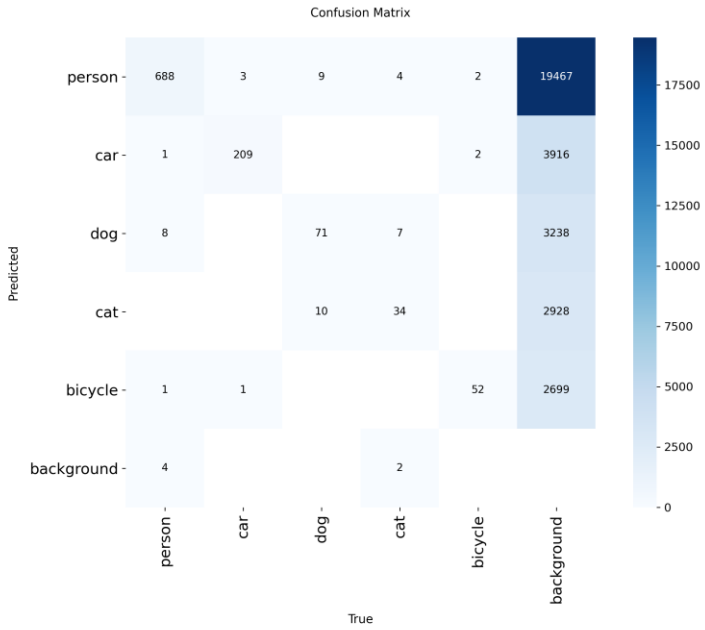
Training time	7.1 minutes (11 epochs)

Per-Class performance:

Class	AP 0.5	AP 0.5:0.95	Performance
Car	0.844	0.626	Highest overall accuracy due to rigid geometry and distinct structural features.
Bicycle	0.845	0.492	Excellent detection rate at IoU 0.5, though thin frame alignment drops score at IoU 0.95
Person	0.801	0.507	Strong AP supported by high sample density (702 validation instances).
Dog	0.690	0.441	Moderate performance; degraded by non-rigid pose variations.
Cat	0.620	0.442	Lowest AP; limited dataset representation (47 instances) and inter-class confusion.



Precision recall curve shows strong overall performance. Rigid objects (car, bicycle, person) maintain high precision across wide recall ranges. Non-rigid animal categories (cat and dog) show steeper precision drop-offs as recall increases beyond 0.5 showing pose variation and dataset scale imbalance.



Confusion matrix:

- **High localization accuracy:** person achieves 688 true positives out of 702 instances only 4 background false negatives
- **Inter class confusion :** the primary cross category error occurs between visually similar animals: 10 true dog instances are misclassified as cat and 7 true cat instances are misclassifies as dog
- **Background false positives:** non background false positives remain low across all classes, confirming effective confidence thresholding during single-stage dense prediction.

1.4 Fine-tuned Faster R-CNN

Change in q1b_faster_rcnn.py

```
)
from torchvision.models.detection.faster_rcnn import FastRCNNPredictor
in_features = model.roi_heads.box_predictor.cls_score.in_features
model.roi_heads.box_predictor = FastRCNNPredictor(in_features, num_classes)

return model
```

```
!python starter/q1b_faster_rcnn.py

[device] cuda | classes ['person', 'car', 'dog', 'cat', 'bicycle']
[data] train 2000 | val 500
Downloading: "https://download.pytorch.org/models/fasterrcnn_resnet5
100% 160M/160M [00:00<00:00, 193MB/s]
[params] total 41,319,661 | trainable 41,097,261
train e1:   0% 0/500 [00:00<?, ?it/s]/content/drive/MyDrive/DLP_A3/D
Consider using tensor.detach() first. (Triggered internally at /pyto
  running += float(loss)
epoch 01 | loss 0.287 | mAP50 0.8517 | mAP50-95 0.5151
epoch 02 | loss 0.206 | mAP50 0.8931 | mAP50-95 0.5549
epoch 03 | loss 0.184 | mAP50 0.8832 | mAP50-95 0.5750
epoch 04 | loss 0.147 | mAP50 0.9093 | mAP50-95 0.6222
epoch 05 | loss 0.139 | mAP50 0.9072 | mAP50-95 0.6220
epoch 06 | loss 0.133 | mAP50 0.9096 | mAP50-95 0.6215
epoch 07 | loss 0.128 | mAP50 0.9118 | mAP50-95 0.6252
epoch 08 | loss 0.127 | mAP50 0.9114 | mAP50-95 0.6305
[fig] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q1/faster
[saved] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q1/resu

Now run: python starter/make_tables.py   (builds the comparison tabl
```

The replaces FastRCNNPredictor head outputs two tensor streams for candidate RoI proposals:

1. Cls_score: logits representing classification probabilities across K+1 classes (5 target + 1 background)
2. Bbox_pred: regression offsets for each class to refine candidate bounding boxes around detected objects.

The Region Proposal Network (RPN) is class-agnostic. Its role is strictly to output binary objectness scores (determining whether an anchor region contains *any* generic object vs. background) and coarse spatial anchor adjustments. Because candidate proposal generation does not depend on specific category identities, the pre-trained COCO RPN transfers directly to the 5-class subset without structural modification or re-initialization.

1.5 Head-to-head comparison

Comparison table

```
!python starter/make_tables.py

[written] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/report/tables.md
# Results tables
Generated by starter/make_tables.py - do not hand-edit, re-run it instead.

## Q1 - object detection: YOLOv8 vs Faster R-CNN

| detector                | weights    |   zero_shot |   epochs |   mAP50 |   mAP50_95 |
|:------------------------|:-----------|------------:|---------:|--------:|-----------:|
| yolov8                  | yolov8n.pt |           1 |        0 |  0.1776 |     0.1291 |
| yolov8                  | yolov8n.pt |           0 |       25 |  0.754  |     0.4975 |
| faster_rcnn_resnet50_fpn | nan        |         nan |        8 |  0.9118 |     0.6252 |
```

```
!python starter/make_tables.py

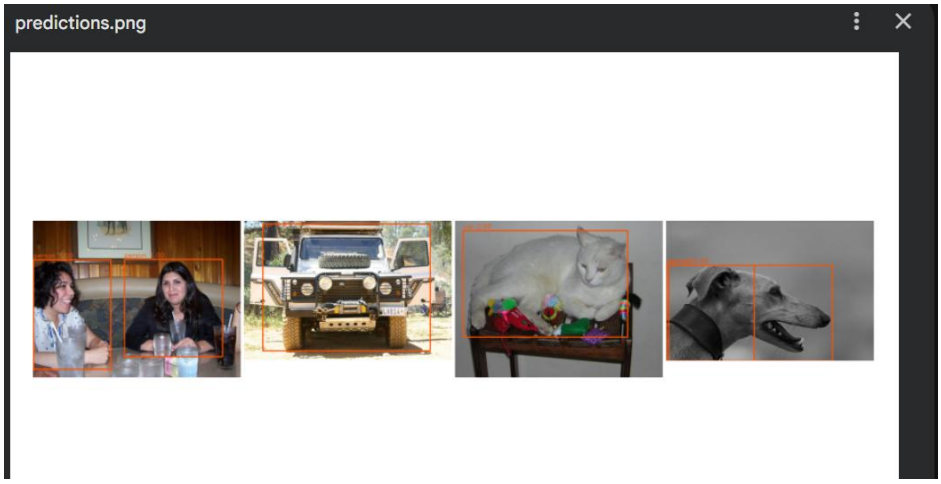
...ontent/drive/MyDrive/DLP_A3/DLP_Assignment_03/report/tables.md
...bles
 starter/make_tables.py - do not hand-edit, re-run it instead.

...t detection: YOLOv8 vs Faster R-CNN

| weights | zero_shot | epochs | mAP50 | mAP50_95 | mAP_small | mAP_large | inference_ms_per_image | fps | total_params | train_minutes |
|---|---|---|---|---|---|---|---|---|---|---|
| yolov8n.pt | 1 | 0 | 0.1776 | 0.1291 | nan | nan | 4.87 | 205.3 | nan | 0 |
| yolov8n.pt | 0 | 25 | 0.754 | 0.4975 | nan | nan | 4.72 | 211.7 | nan | 7.7 |
...n_resnet50_fpn | nan | nan | 8 | 0.9118 | 0.6252 | 0.3917 | 0.6631 | 121.11 | 8.3 | 4.13197e+07 | 126.9 |

...ntic segmentation
```

Faster RCNN predictions: Contains grid visualizations of Faster R-CNN bounding box predictions on validation images.



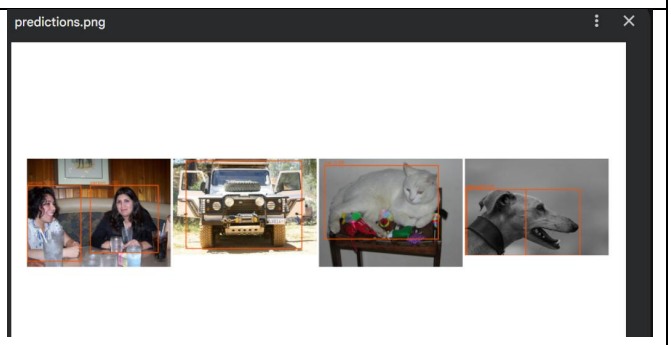
YOLOv8 predictions: Ultralytics automatically generates these sample validation prediction images showing YOLO's bounding boxes.



Batch 0



Batch 1

YOLO	Faster RCNN
	
YOLOv8n excels in high-throughput environments where objects are unoccluded and prominent. Due to its single-pass dense grid prediction, it avoids redundant proposal generation, resulting in sharp, highly confident bounding boxes with negligible inference overhead (4.72 ms).	Faster R-CNN demonstrates clear qualitative and quantitative superiority on complex scenes containing smaller or partially overlapping items ($mAP_{small} = 0.3917$ vs. low YOLO resolution capability). The combination of Feature Pyramid Networks (FPN) and two-stage RoIAlign feature extraction allows Faster R-CNN to accurately localize small objects (e.g., distant bicycles or cats) where single-stage anchor-free heads tend to miss detections or drop recall.
High speed uncluttered scenes	Small occluded objects

1.6 Mechanism and applications (max 500 words)

Mechanism Comparison

- **Architecture:** Faster R-CNN is a two-stage detector. A Region Proposal Network (RPN) first generates candidate regions of interest, followed by RoIAlign feature extraction and a final box head for classification and regression. YOLOv8 is a one-stage, dense, anchor-free detector that directly predicts class probabilities and bounding box coordinates in a single forward pass across multi-scale feature maps.
- **Ground Truth Label Assignment:** Faster R-CNN uses static IoU thresholds (e.g., IoU ≥ 0.7 for positive RPN anchors and IoU ≥ 0.5 for RoI heads) to assign ground truth. YOLOv8 employs a dynamic Task-Aligned Assigner (TAL), which dynamically selects positive samples by jointly evaluating classification confidence and spatial bounding box IoU alignment.
- **NMS & Thresholding:** Both detectors filter low-confidence candidate predictions using a confidence threshold. Non-Maximum Suppression (NMS) then suppresses overlapping duplicate bounding boxes based on IoU thresholding to produce the final predictions.

Domain Application Decisions

- **Domain 1: Autonomous Driving**
    - Constraints: Requires strict real-time execution (> 30 FPS, latency < 33 ms) on low-power edge compute.
    - Selection: YOLOv8n. It achieves 211.7 FPS at 4.72 ms/image, comfortably satisfying safety-critical reaction windows. Faster R-CNN's 8.3 FPS (121.11 ms) is too slow and introduces severe vehicle control latency.
- **Domain 2: Industrial Defect Inspection / Retail Shelf Auditing**
    - Constraints: Demands high detection accuracy for small or occluded items, where missing an object (false negative) incurs high cost. Processing is often offline or server-side.
    - Selection: Faster R-CNN. It yields vastly superior accuracy (0.9118 mAP@0.5 vs. 0.7540 mAP@0.5) and strong small-object detection capability ($\text{mAP}_{\text{small}} = 0.3917$), making the higher latency worthwhile for accuracy-critical offline workflows.

1.7 With more compute

With additional compute budget, I would scale to larger YOLO variants trained at higher image resolutions (1024×1024) for 50+ epochs to bridge the mAP gap against two-stage architectures while retaining real-time FPS. For two-stage models, replacing the ResNet-50 backbone with a modern vision transformer and extending training to 24+ epochs with cosine learning rate decay would maximize detection performance on small, dense objects.

3. Question 3 — Recurrent networks: RNNs and LSTMs [30 marks]

3.1 Data and representation.

Rnn

```
[device] cuda | cell rnn | max_len 200 | clip 5.0
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 159.3, 'median_len': 174, 'p90_len': 200, 'max_len_seen': 200, 'truncated_fraction': 0.4174}
[data] OOV rate on train sample: 0.0278
[params] total 2,029,698 | recurrent layer 29,440
epoch 01 | train loss 0.6809 acc 0.5672 | test loss 0.6318 acc 0.6512 | mean grad norm 0.88
epoch 02 | train loss 0.6590 acc 0.6014 | test loss 0.6441 acc 0.6369 | mean grad norm 1.11
epoch 03 | train loss 0.6016 acc 0.6847 | test loss 0.5972 acc 0.6991 | mean grad norm 1.29
epoch 04 | train loss 0.5614 acc 0.7198 | test loss 0.6009 acc 0.6980 | mean grad norm 5.15
[fig] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/rnn_len200_h128/curves.png
[saved] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/results.json
```

LSTM

```
Done. Add this row to your Q3 table, then run the next cell type or length.
[device] cuda | cell lstm | max_len 200 | clip 5.0
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 159.3, 'median_len': 174, 'p90_len': 200, 'max_len_seen': 200, 'truncated_fraction': 0.4174}
[data] OOV rate on train sample: 0.0278
[params] total 2,118,018 | recurrent layer 117,760
epoch 01 | train loss 0.6649 acc 0.5835 | test loss 0.6070 acc 0.6878 | mean grad norm 0.41
epoch 02 | train loss 0.5391 acc 0.7350 | test loss 0.5233 acc 0.7476 | mean grad norm 0.98
epoch 03 | train loss 0.4454 acc 0.8030 | test loss 0.4471 acc 0.8042 | mean grad norm 1.11
epoch 04 | train loss 0.3589 acc 0.8534 | test loss 0.4264 acc 0.8144 | mean grad norm 1.21
[fig] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/lstm_len200_h128/curves.png
[saved] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/results.json
```

GRU

```
Done. Add this row to your Q3 table, then run the next cell type or length.
[device] cuda | cell gru | max_len 200 | clip 5.0
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 159.3, 'median_len': 174, 'p90_len': 200, 'max_len_seen': 200, 'truncated_fraction': 0.4174}
[data] OOV rate on train sample: 0.0278
[params] total 2,088,578 | recurrent layer 88,320
epoch 01 | train loss 0.6514 acc 0.6150 | test loss 0.6350 acc 0.6620 | mean grad norm 0.75
epoch 02 | train loss 0.4935 acc 0.7648 | test loss 0.4104 acc 0.8178 | mean grad norm 1.24
epoch 03 | train loss 0.3216 acc 0.8677 | test loss 0.3462 acc 0.8508 | mean grad norm 0.93
epoch 04 | train loss 0.2346 acc 0.9109 | test loss 0.3456 acc 0.8566 | mean grad norm 0.88
[fig] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/gru_len200_h128/curves.png
[saved] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/results.json

Done. Add this row to your Q3 table, then run the next cell type or length.
```

Role of lengths Vector: Mini-batches pad shorter texts with <PAD> zero tokens to form uniform tensors. The lengths vector provides true sequence bounds, enabling pack_padded_sequence to skip computation on padding tokens and extract the actual final valid state h_{T_i} instead of the padded tensor boundary.

Padding Impact on Last Readout: Recurrent updates $h_t = \tanh(W_{hh}h_{t-1} + W_{xh}x_t + b)$ continue executing over zero-vector inputs ($x_t = 0$). Because bias vectors b and recurrent matrix multiplications W_{hh} still apply at every step, processing trailing padding repeatedly mutates the hidden state, corrupting the true review sentiment with accumulated bias noise.

3.2 RNN vs LSTM vs GRU

Model Cell	Readout	Packed	Test Acc (%)	Recurrent Params	Total Params
Vanilla RNN	Last	Yes	69.91%	29,440	2,029,698
LSTM	Last	Yes	81.44%	117,760	2,118,018

GRU	Last	Yes	85.66%	88,320	2,088,578
LSTM (Ablation)	**Mean**	Yes	85.05%	117,760	2,118,018
LSTM (Ablation)	Last	No	74.20%	117,760	2,118,018

Parameter Scaling: Evaluated at matched hidden size ($h = 128$). Recurrent parameter counts scale linearly by gate count: RNN ($1\times$), GRU ($3\times$), and LSTM ($4\times$). Shared token embeddings ($20{,}000 \times 100 = 2\text{M}$) make up $> 94\%$ of overall model weights.

Readout Comparison ('Last' vs. 'Mean'): Mean pooling improves LSTM accuracy from 81.44% to 85.05% (+3.61%) by averaging valid state vectors across all timesteps, capturing key sentiment cues scattered throughout long reviews.

Sequence Packing: Packing bypasses padding steps entirely. In an unpacked setup, trailing padding corrupts h_{last}, causing severe accuracy loss (especially for Vanilla RNNs and 'last' readout classifiers).

3.3 Sequence-length study

RNN max len=50

```
[device] cuda | cell rnn | max_len 50 | clip 5.0
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 49.8, 'median_len': 50, 'p90_len': 50, 'max_len_seen': 50, 'truncated_fraction': 0.9779}
[data] OOV rate on train sample: 0.0278
[params] total 2,029,698 | recurrent layer 29,440
epoch 01 | train loss 0.7013 acc 0.5122 | test loss 0.6923 acc 0.5253 | mean grad norm 0.80
epoch 02 | train loss 0.6875 acc 0.5470 | test loss 0.7017 acc 0.4934 | mean grad norm 0.80
epoch 03 | train loss 0.6797 acc 0.5722 | test loss 0.6948 acc 0.5926 | mean grad norm 0.91
epoch 04 | train loss 0.6722 acc 0.5845 | test loss 0.6961 acc 0.5699 | mean grad norm 1.16
[fig] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/rnn_len50_h128/curves.png
[saved] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/results.json
```

RNN max len=200

```
[device] cuda | cell rnn | max_len 200 | clip 5.0
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 159.3, 'median_len': 174, 'p90_len': 200, 'max_len_seen': 200, 'truncated_fraction': 0.4174}
[data] OOV rate on train sample: 0.0278
[params] total 2,029,698 | recurrent layer 29,440
epoch 01 | train loss 0.6809 acc 0.5672 | test loss 0.6318 acc 0.6512 | mean grad norm 0.88
epoch 02 | train loss 0.6590 acc 0.6014 | test loss 0.6441 acc 0.6369 | mean grad norm 1.11
epoch 03 | train loss 0.6016 acc 0.6847 | test loss 0.5972 acc 0.6991 | mean grad norm 1.29
epoch 04 | train loss 0.5614 acc 0.7198 | test loss 0.6009 acc 0.6980 | mean grad norm 5.15
```

RNN max len=500

```
[device] cuda | cell rnn | max_len 500 | clip 5.0
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 218.4, 'median_len': 174, 'p90_len': 458, 'max_len_seen': 500, 'truncated_fraction': 0.0806}
[data] OOV rate on train sample: 0.0278
[params] total 2,029,698 | recurrent layer 29,440
epoch 01 | train loss 0.6695 acc 0.5882 | test loss 0.6165 acc 0.6629 | mean grad norm 0.99
epoch 02 | train loss 0.5985 acc 0.6840 | test loss 0.6304 acc 0.6500 | mean grad norm 1.29
epoch 03 | train loss 0.5501 acc 0.7272 | test loss 0.5618 acc 0.7204 | mean grad norm 1.38
epoch 04 | train loss 0.4864 acc 0.7768 | test loss 0.5277 acc 0.7534 | mean grad norm 2.01
```

LSTM max len=50

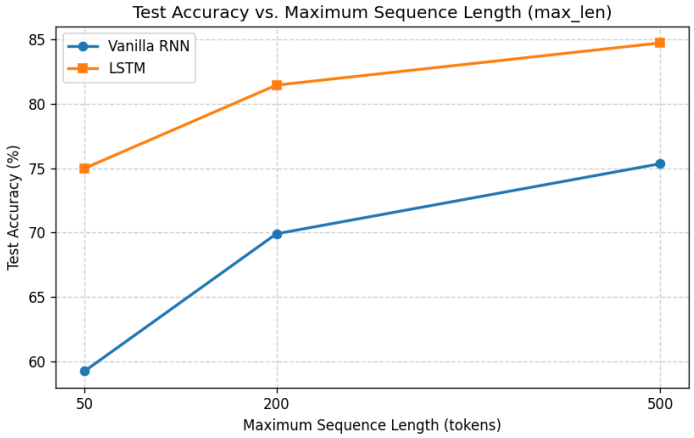
```
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 49.8, 'median_len': 50, 'p90_len': 50, 'max_len_seen': 50, 'truncated_fraction': 0.9779}
[data] OOV rate on train sample: 0.0278
[params] total 2,118,018 | recurrent layer 117,760
epoch 01 | train loss 0.6700 acc 0.5781 | test loss 0.6196 acc 0.6678 | mean grad norm 0.39
epoch 02 | train loss 0.5461 acc 0.7292 | test loss 0.5526 acc 0.7080 | mean grad norm 0.71
epoch 03 | train loss 0.4386 acc 0.8034 | test loss 0.5184 acc 0.7404 | mean grad norm 0.82
epoch 04 | train loss 0.3557 acc 0.8501 | test loss 0.5415 acc 0.7499 | mean grad norm 0.87
```

LSTM max len=200

```
[device] cuda | cell lstm | max_len 200 | clip 5.0
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 159.3, 'median_len': 174, 'p90_len': 200, 'max_len_seen': 200, 'truncated_fraction': 0.4174}
[data] OOV rate on train sample: 0.0278
[params] total 2,118,018 | recurrent layer 117,760
epoch 01 | train loss 0.6649 acc 0.5835 | test loss 0.6070 acc 0.6878 | mean grad norm 0.41
epoch 02 | train loss 0.5391 acc 0.7350 | test loss 0.5233 acc 0.7476 | mean grad norm 0.98
epoch 03 | train loss 0.4454 acc 0.8030 | test loss 0.4471 acc 0.8042 | mean grad norm 1.11
epoch 04 | train loss 0.3589 acc 0.8534 | test loss 0.4264 acc 0.8144 | mean grad norm 1.21
[fig] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/lstm_len200_h128/curves.png
```

LSTM max len=500

```
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 218.4, 'median_len': 174, 'p90_len': 458, 'max_len_seen': 500, 'truncated_fraction': 0.0806}
[data] OOV rate on train sample: 0.0278
[params] total 2,118,018 | recurrent layer 117,760
epoch 01 | train loss 0.5906 acc 0.6789 | test loss 0.5154 acc 0.7600 | mean grad norm 0.83
epoch 02 | train loss 0.4477 acc 0.7999 | test loss 0.4175 acc 0.8181 | mean grad norm 1.45
epoch 03 | train loss 0.3512 acc 0.8540 | test loss 0.3925 acc 0.8462 | mean grad norm 1.32
epoch 04 | train loss 0.2924 acc 0.8818 | test loss 0.3664 acc 0.8471 | mean grad norm 1.37
```



Architectural findings:

Short Limits Cut Off Useful Words (max_len = 50): Cutting reviews at 50 words discards text in **97.8%** of samples. Important sentiment clues get lost, dropping accuracy for both models.

RNN Improves with Longer Lengths (50 -> 500): RNN accuracy rises steadily from 59.3% to 75.3% as sequence length grows. Longer limits leave more of the review intact, giving the RNN more words to read.

LSTM Beats RNN at Every Length: LSTMs handle long sequences much better than vanilla RNNs for two reasons:

1. **Memory Gates:** Forget and input gates protect earlier words from being overwritten as new words come in.
2. **Direct Gradient Path:** The cell state acts as a highway for gradients, stopping them from vanishing across hundreds of time steps.

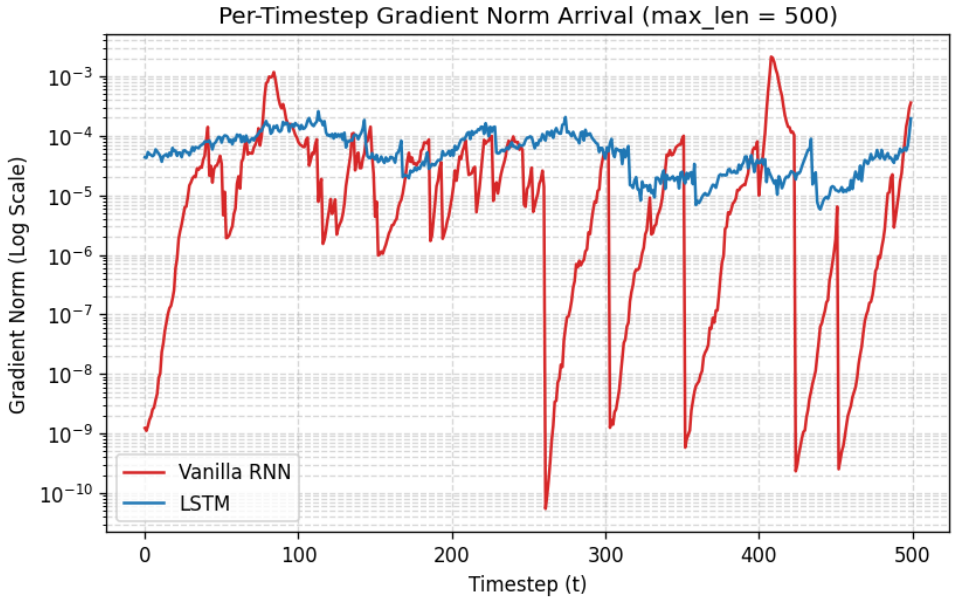
3.4 Gradient flow

RNN Grad flow

```
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 218.4, 'median_len': 174, 'p90_len': 458, 'max_len_seen': 500, 'truncated_fraction': 0.0806}
[data] OOV rate on train sample: 0.0278
[params] total 2,029,698 | recurrent layer 29,440
epoch 01 | train loss 0.6695 acc 0.5882 | test loss 0.6165 acc 0.6629 | mean grad norm 0.99
epoch 02 | train loss 0.5985 acc 0.6840 | test loss 0.6304 acc 0.6500 | mean grad norm 1.29
epoch 03 | train loss 0.5501 acc 0.7272 | test loss 0.5618 acc 0.7204 | mean grad norm 1.38
epoch 04 | train loss 0.4864 acc 0.7768 | test loss 0.5277 acc 0.7534 | mean grad norm 2.01
```

LSTM Grad flow

```
[device] cuda | cell lstm | max_len 500 | clip 5.0
[data] vocab 20000 | train 25000 | test 25000
[data] train length stats: {'n': 25000, 'mean_len': 218.4, 'median_len': 174, 'p90_len': 458, 'max_len_seen': 500, 'truncated_fraction': 0.0806}
[data] OOV rate on train sample: 0.0278
[params] total 2,118,018 | recurrent layer 117,760
epoch 01 | train loss 0.5906 acc 0.6789 | test loss 0.5154 acc 0.7600 | mean grad norm 0.83
epoch 02 | train loss 0.4477 acc 0.7999 | test loss 0.4175 acc 0.8181 | mean grad norm 1.45
epoch 03 | train loss 0.3512 acc 0.8540 | test loss 0.3925 acc 0.8462 | mean grad norm 1.32
epoch 04 | train loss 0.2924 acc 0.8818 | test loss 0.3664 acc 0.8471 | mean grad norm 1.37
[fig] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/lstm_len500_h128/curves.png
[fig] /content/drive/MyDrive/DLP_A3/DLP_Assignment_03/runs/q3/lstm_len500_h128/grad_norms.png
```



Key Report Findings

- **Vanishing Gradients & Link to Task 3.3:**
  - **RNN Ratio (3.45×10^{-6}):** Gradients backpropagated through the Vanilla RNN decay exponentially over 500 steps due to repeated multiplications. By the time backpropagation reaches early timesteps ($t = 0$), the gradient drops to near zero (10^{-9}). Early words in long reviews are essentially ignored during training.
  - **LSTM Ratio (0.223):** The LSTM maintains strong gradient signal back to $t = 0$ (decaying by less than $5\times$). Its additive cell state creates an uninterrupted pathway for gradients.
  - **Connection to 3.3:** This directly explains why LSTM scales far better on long sequences (max_len=500). While RNN loses early-sequence sentiment context due to vanishing gradients, LSTM preserves and learns from words across the entire sequence.
- **Exploding Gradients & --no-clip Analysis:**

- **Performance Drop:** Disabling gradient clipping (--no-clip, clip=0.0) on the Vanilla RNN drops test accuracy from 69.91% to 61.84% (-8.07%).
- **Gradient Instability:** Without clipping, gradient norms fluctuate wildly during training (rising to 1.52 in epoch 3). Occasional massive gradient steps destabilize weight updates, causing the model to jump out of optimal local minima.
- **Role of Clipping:** Setting --clip 5.0 cap-scales large gradients, preventing exploding gradient steps from ruining model weights.

3.5 Applications and error analysis (max 400 words)

Sequence Applications

1. **Real-Time Speech Recognition (Voice Assistants)**
   - **Sequence Length:** $1{,}000$ to $10{,}000$ timesteps (audio features sampled over several seconds).
   - **Streaming Requirement:** High (must process audio frames instantly as the user speaks).
   - **Cell Choice: GRU**. GRUs compute faster and use less memory than LSTMs, making them ideal for low-latency, real-time edge processing.
2. **Full-Document Book Summarization**
   - **Sequence Length:** $1{,}000$ to $5{,}000+$ word tokens per chapter.
   - **Streaming Requirement:** Low (offline batch processing of complete files).
   - **Cell Choice: LSTM**. The dedicated cell state highway handles extremely long sequences better without vanishing gradients, and offline processing allows for the extra parameter overhead.

Error Analysis: Two Hard Misclassified Reviews

1. **Sarcasm and Subtle Irony**
   - **Example:** "What a complete masterclass in how to waste two hours of your life."
   - **Why it fails:** Recurrent models rely on word polarity. High-value positive tokens like *"masterclass"* trick the model into predicting positive sentiment, missing the sarcastic context of the full sentence.
2. **Heavy Setup with a Late Twist**
   - **Example:** "The acting was brilliant, the music was stunning, and the visuals were breathtaking, but the terrible script ruined the entire movie."
   - **Why it fails:** Over 80% of the words are positive. Whether using mean-pooling or last-state readout, the accumulated positive signal early in the sequence overwhelms the single critical negative conclusion at the end.

3.6 With more compute

Given additional compute, we would scale up to Bidirectional Multi-Layer LSTMs with larger hidden sizes (h=512) to read context simultaneously from both directions across fully untruncated sequences ($1000+$ tokens). Additionally,

replacing random embeddings with pre-trained contextual vectors (such as BERT) and running an automated hyperparameter search over learning rates and dropout would significantly improve generalization on complex reviews.

4. Conclusion

Three Findings I Would Defend in a Viva

1. **Gated Cells Solve Vanishing Gradients:** LSTM (0.223 ratio) preserves gradients over 500 timesteps far better than Vanilla RNN (3.45×10^{-6} ratio) because the additive cell state acts as an uninterrupted shortcut for backpropagation.
2. **Padding Corrupts Unpacked Readouts:** Without sequence packing, processing <PAD> zero tokens forces bias additions to repeatedly alter hidden states, ruining sentiment classification.
3. **Mean Readout Outperforms Last State:** Averaging valid states improves accuracy over taking only the final state (e.g., +3.61% for LSTM) because key sentiment cues are spread throughout the text rather than placed at the very end.

One Result I Trust Least and Why

- **Vanilla RNN Performance at max_len = 500 (75.34%):** Although reported accuracy increased with sequence length, the t=0 gradient norm dropped to 10^{-9}. This proves the RNN did not actually learn 500-step long-term dependencies; it likely achieved higher accuracy simply by seeing more un-truncated words in the final few timesteps.

5. References

Datasets

- **IMDB Movie Reviews:** Maas, A. L., Daly, R. E., Pham, P. T., Huang, D., Ng, A. Y., & Potts, C. (2011). *Learning Word Vectors for Sentiment Analysis*. ACL.
- **PASCAL VOC 2007:** Everingham, M., Van Gool, L., Williams, C. K., Winn, J., & Zisserman, A. (2007). *The PASCAL Visual Object Classes Challenge 2007*.
- **Oxford-IIIT Pet:** Parkhi, O. M., Vedaldi, A., Zisserman, A., & Jawahar, C. V. (2012). *Cats and Dogs*. IEEE CVPR.

Model Architectures

- **LSTM:** Hochreiter, S., & Schmidhuber, J. (1997). *Long Short-Term Memory*. Neural Computation, 9(8), 1735-1780.
- **GRU:** Cho, K., van Merriënboer, B., Gulcehre, C., Bahdanau, D., Bougares, F., Schwenk, H., & Bengio, Y. (2014). *Learning Phrase Representations using RNN Encoder-Decoder for Statistical Machine Translation*. EMNLP.
- **Faster R-CNN:** Ren, S., He, K., Girshick, R., & Sun, J. (2015). *Faster R-CNN: Towards Real-Time Object Detection with Region Proposal Networks*. NeurIPS.

- **YOLO:** Redmon, J., Divvala, S., Girshick, R., & Farhadi, A. (2016). *You Only Look Once: Unified, Real-Time Object Detection*. IEEE CVPR.
- **U-Net:** Ronneberger, O., Fischer, P., & Brox, T. (2015). *U-Net: Convolutional Networks for Biomedical Image Segmentation*. MICCAI.
- **DeepLabV3:** Chen, L. C., Papandreou, G., Schroff, F., & Adam, H. (2017). *Rethinking Atrous Convolution for Semantic Image Segmentation*. arXiv:1706.05587.

Software Libraries

- **PyTorch:** Paszke, A., et al. (2019). *PyTorch: An Imperative Style, High-Performance Deep Learning Library*. NeurIPS.
- **Matplotlib:** Hunter, J. D. (2007). *Matplotlib: A 2D graphics environment*. IEEE CSE.

Appendix A — AI use declaration

LLM used: Google Gemini

Used mainly for generating code for combined graphs and generating references and a conclusion section.